Informatica®

Informatica® Cloud Data Integration

Pinecone Connector

Informatica Cloud Data Integration Pinecone Connector
October 2025

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Preface

Use Pinecone Connector to learn how to write to Pinecone vector database by using Data Integration. Learn to create a Pinecone connection, run mappings in advanced mode and tasks in Data Integration.

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CHAPTER 1

Introduction to Pinecone Connector

You can use Pinecone Connector to write unstructured text files, such as PDFs, as vector embeddings along with their associated metadata to the Pinecone vector database.

You can use a Pinecone connection as a target in mappings in advanced mode. When you run a mapping in advanced mode or a task based on a mapping in advanced mode, Data Integration analyzes the data logic in the mapping to automatically assign data logic to run on an advanced cluster.

Example

You run a legal firm that handles a massive repository of legal documents, contracts, and case files in PDF format in Microsoft Azure Data Lake Storage Gen2.

Legal professionals need efficient ways to search and retrieve relevant documents quickly, often using complex queries that go beyond simple keyword searches.

To enhance the speed and accuracy of document retrieval, you want to integrate your data into the Pinecone vector database. This integration allows legal professionals to efficiently query high-dimensional vectors that represent complex data, such as text and tables, streamlining their search processes and enhancing overall productivity.

To integrate data in Pinecone, you perform the following steps:

1. Create a mapping in advanced mode.
2. Add a Source transformation to read data from Microsoft Azure Data Lake Storage Gen2.
3. Add a Chunking transformation to split the text into smaller manageable chunks for enhanced vector representation.
4. Add an Embedding transformation to convert the text to vector embeddings.
5. Add a Target transformation to use the Pinecone connection and load the vector embeddings with their associated metadata to Pinecone.
6. Run the mapping in advanced mode.

Pinecone Connector assets

Create assets in Data Integration to integrate data using Pinecone Connector.

When you use Pinecone Connector, you can include the following Data Integration assets:

- Mappings in advanced mode
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for Pinecone

Create a Pinecone connection to securely write data to Pinecone database.

You can use a Pinecone connection to specify targets in mappings in advanced mode and mapping tasks.

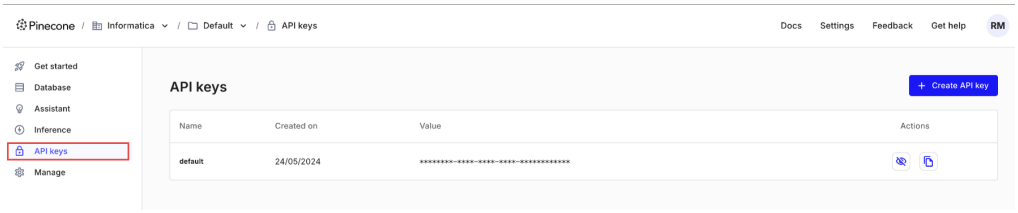
Prepare for authentication

Before you configure the connection properties, you need to get the API key of your Pinecone account to authenticate access to the Pinecone APIs.

Get the Pinecone API key

You need an API key to make API calls to your Pinecone project.
To get the Pinecone API key, perform the following steps:

1. Open the Pinecone console.
2. Select your project.
3. Go to **API Keys**.



4. Copy the API key.

Connect to Pinecone

Let's configure the Pinecone connection properties to connect to Pinecone.

Before you begin

Before you get started, you'll need to get the API key from your Pinecone account.

Check out "Prepare for authentication" on page 8 to learn about the authentication requirements before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure agent or serverless runtime environment. Hosted Agent is not applicable for mappings in advanced mode.
API Key	The API key of your Pinecone account to authenticate access to the Pinecone APIs.

CHAPTER 3

Mappings for Pinecone

You can configure a mapping in advanced mode to write to Pinecone.

Use the Data Integration Mapping Designer to define and configure a mapping in advanced mode for Pinecone target. In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

You can configure a Target transformation to represent a Pinecone index. Add the Target transformations in the mapping canvas and configure the Pinecone target properties. When you run a mapping in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

Prerequisites

Before you write data to a Pinecone database, ensure that your Pinecone project includes a pre-created index with the correct dimensionality based on your vector embedding technique.

You can create either a serverless or pod-based index. The indexes can be hosted on cloud platforms such as Amazon, Azure, or Google Cloud Platform.

For more information on how to create a Pinecone index, see the Pinecone documentation.

Targets properties for Pinecone

Add a Target transformation to write data to a target. When you add a Target transformation to a mapping in advanced mode, you define the target connection, target objects, and target properties related to the Pinecone connection type.

The following table describes the Pinecone target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection.
Target Type	Target type. Select **Single Object**.
Object	Name of the target object that represents an index in the Pinecone project.

Property	Description
Operation	Target operation. You can use only the upsert operation.
Update Columns	The id field of the Pinecone index that is used as the primary key column when you upsert target data.

The following table describes the advanced properties:

Property	Description
Batch Size	The number of vectors that you can upsert into the Pinecone index in a single API request. Default is 100. Maximum value is 1000.
Namespace	The namespace in the selected index where you want to upsert vectors. If you do not specify a namespace, the vectors are saved in the Default namespace in the index. Maximum value is 512 ASCII printable characters.

The following table describes the target fields that a Target transformation writes to Pinecone:

Property	Description
id	Required. Unique identifier for each record. You can use a Sequence Generator or Expression transformation to create vector IDs. Maximum value is 512 characters.
values	Required. A dense vector or a vector embedding which is a series of numerical values that represent different dimensions of the data that are essential for understanding patterns, relationships, and underlying structures such as semantic information. You can use a Vector Embedding transformation to create vector embeddings. Maximum value is 20000 dimensions.
metadata	Optional. Metadata to write to a Pinecone index. Metadata is the additional information associated with the vector embeddings. For example, the original text of the embeddings can be stored in the metadata. Maximum value of metadata per record is 40 KB.

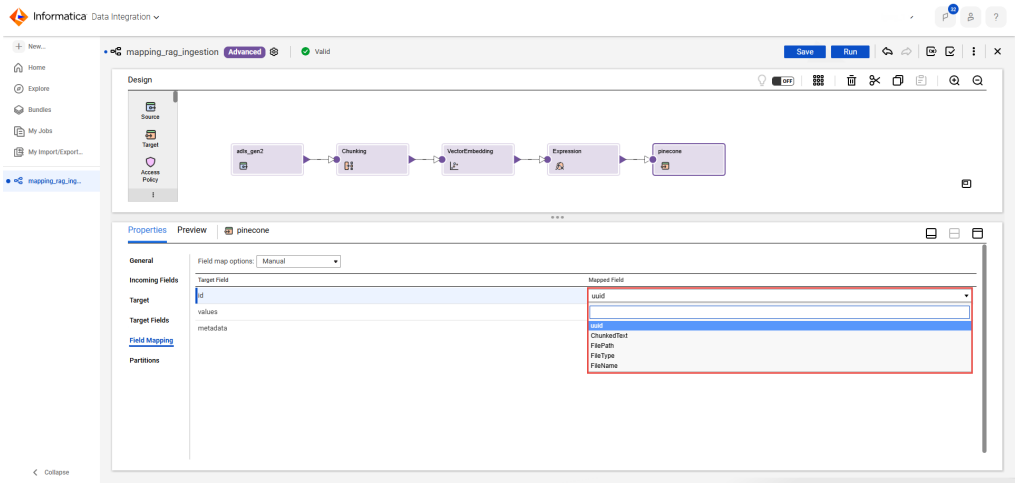
For more information about configuring transformations, see *Transformations* in the Data Integration documentation.

Field mapping

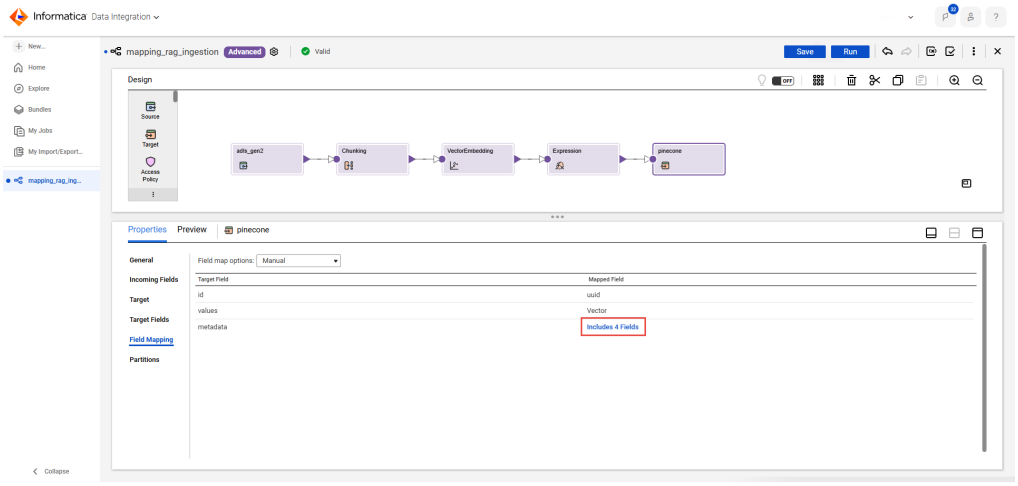
When you configure a Pinecone target, you can map the target fields in Pinecone to the appropriate incoming fields from the upstream transformation and select the metadata you want to upsert into Pinecone.

To configure the field mapping, perform the following steps:

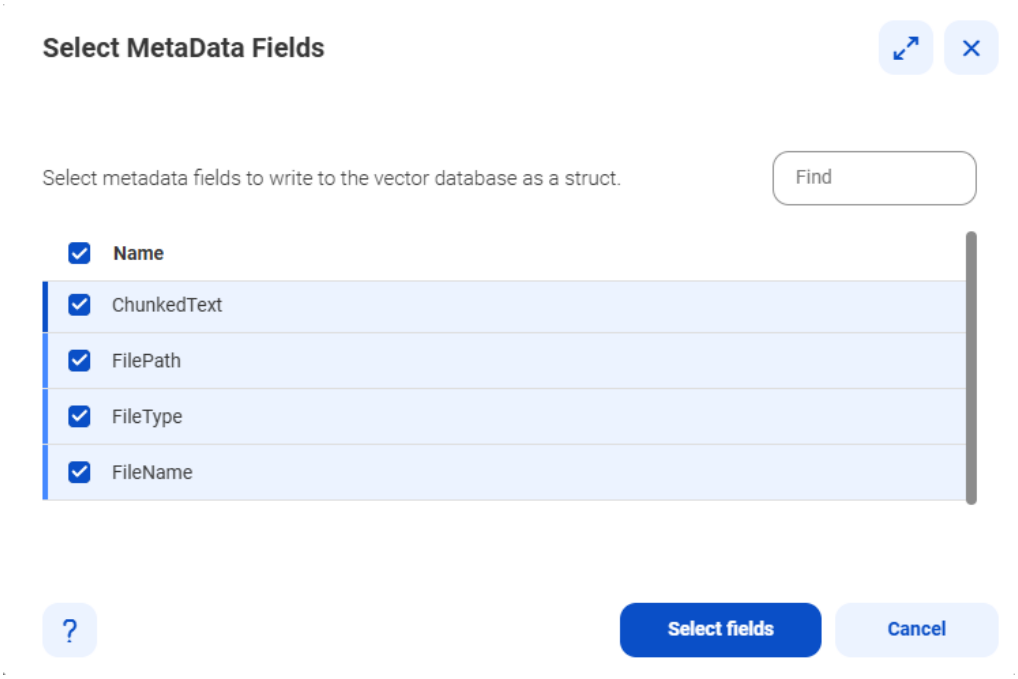
1. Go to the **Field Mapping** tab.
2. To map the **id** and **values** fields, under **Mapped Field**, select the appropriate incoming fields from the list.



3. To select the metadata fields that you want to upsert into Pinecone, click the link under **Mapped Field**.



4. Select the metadata fields, and then click **Select Fields**.



Rules and guidelines for Pinecone Connector

Consider the following rules and guidelines when you use Pinecone Connector:

- The maximum batch size must not exceed 2 MB per batch. You can configure the batch size in advanced target properties.
- The maximum metadata size per record must not exceed 40 KB. If the size exceeds 40 KB, chunk the text further to reduce the metadata size.
- The maximum length for a record ID must not exceed 512 characters.
- Dense vectors must contain at least one non-zero value.
- Metadata must not contain null values. Instead of setting a key to hold a null value, remove the key from the metadata payload.

For more information about the rules and guidelines for the Pinecone APIs, see the Pinecone documentation.

CHAPTER 4

Data type reference

Data Integration uses the following data types in Pinecone mappings in advanced mode:

The following table lists the Pinecone data types that are compatible with Data Integration and the corresponding transformation data types:

- Pinecone native data types that are inherent to the Pinecone database.
- Transformation data types. Set of data types that appear in the transformations. These are internal data types based on ANSI SQL-92 generic data types, which the agent uses to move data across platforms. They appear in all transformations in a mapping in advanced mode.

When the agent writes to a Pinecone target, it converts the transformation data types to the comparable native data types.

Pinecone and transformation data types

The following table lists the Pinecone data types that Data Integration supports and the corresponding transformation data types:

Pinecone data type	Transformation data type	Range and Description
String	String	The maximum value for the id field is 512 characters. The maximum value for the metadata per record in 40 KB.
Vector	Vector	Range of dense vector or the values field is 1 to 20000 dimensions.

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